MILITARY SPECIFICATION

POWER SUPPLY, VARIABLE OUTPUT (0 to 50 VOLTS, 0 to 5 AMPERES)

1. SCOPE

1.1 This specification covers one type of power supply used in Navy electronic equipments. The power supply shall be capable of an output voltage of 0 to 50 volts direct current (Vdc) at 0 to 5 amperes with a primary potential of 105 to 125 volts and a frequency of 60 Hertz (Hz) ± 5 percent.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-8625 - Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories and Repair Parts: Packaging and Packing of.

STANDARDS

MILITARY

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-275 - Printed Wiring For Electronic Equipment.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-701 - Lists of Standard Semiconductor Devices.
MIL-STD-1285 - Marking of Electrical and Electronic Parts.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article sample. Prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.3).

3.2 Materials. The materials shall be as specified herein; however, when a definite material is not specified a material shall be used which will enable the power supply to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

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3.3 Semiconductor devices. Semiconductor devices shall be selected in accordance with MIL-STD-701.

3.4 Printed wiring boards. Printed wiring boards shall be in accordance with MIL-STD-275.

3.5 Environmental

3.5.1 Shock. After being tested as specified in 4.5.1.1 the power supply shall not have been damaged or adversely affected, and shall meet the requirements of 3.6.2 through 3.6.5.

3.5.2 Vibration. When tested as specified in 4.5.1.2, the output voltage shall be monitored and shall not vary by more than 5 percent. Upon completion of the test the power supply shall meet the requirements of 3.6.2 through 3.6.5.

3.5.3 Temperature.

3.5.3.1 Maximum operating temperature (heat run). The maximum operating temperature of the power supply shall not exceed 105° C under any variations of the frequency and input voltage (see 3.6.1) and at full dc load of 50 volts and 5 amperes. When tested as specified in 4.5.1.3.1 the power supply shall conform to the requirements of 3.6.2 through 3.6.5.

3.5.3.2 Low temperature operation. After being tested as specified in 4.5.1.3.2 the power supply shall meet the requirements of 3.6.2 through 3.6.5.

3.5.3.3 Storage temperature. When tested as specified in 4.5.1.3.3 the power supply shall meet the requirements of 3.6.2 through 3.6.5.

3.6 Electrical.

3.6.1 Frequency. The power supply shall be operated on a frequency of 60 Hz ± 5 percent with an input voltage of 105 to 125 Vac.

3.6.2 Output voltage. The output voltage shall be capable of being adjusted from 0 to 50 Vdc. The output voltage shall be within ± 2 percent of the value indicated on the panel voltmeter when tested in accordance with 4.5.2.1.

3.6.3 Output current. The output current shall be capable of being adjusted from 0 to 5 amperes dc. The output current shall be within ± 2 percent of the value indicated on the panel ammeter when tested as specified in 4.5.2.1.

3.6.4 Power drain. When tested in accordance with 4.5.2.1 the primary power to the power supply shall not exceed 2 amperes ac when operating under full dc load of 50 volts and 5 amperes.

3.6.5 Line regulation. Line regulation shall be 0.01 percent + 1 millivolt (mV) with an ac line variation of 105 to 125 Vac when tested in accordance with 4.5.2.5.

3.6.6 Stability. Stability shall be better than 0.02 percent + 3 mV for 24 hours at constant line load and ambient temperature after warm-up when tested in accordance with 4.5.2.6.

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3.6.7 Temperature coefficient of output voltage. Temperature coefficient of output voltage shall be less than .02 percent per degree Centigrade when tested as specified in 4.5.2.7.

3.6.8 Voltage regulation. Voltage regulation shall be .01 percent maximum when tested as specified in 4.5.2.2.

3.6.9 Voltage ripple. Voltage ripple shall be 0.5 mV maximum when tested as specified in 4.5.2.3.

3.6.10 Output impedance. The output impedance shall be .004 ohms maximum when tested as specified in 4.5.2.4.

3.6.11 Overload. The power supply shall be capable of operating into a short circuit without damage when tested as specified in 4.5.2.10. Subsequent to the test the power supply shall not be damaged and shall meet the requirements of 3.6.2 through 3.6.5. A blown fuse is not considered damage or failure.

3.6.12 Terminal grounding. Either the positive or negative dc terminal may be grounded or operated at 200 Vdc from ground.

3.6.13 Run-in. The power supply shall be subjected to a 48 hour run-in test in accordance with 4.5.2.8. Subsequent to the test the power supply shall not be damaged and shall meet the requirements of 3.6.2 through 3.6.5.

3.6.14 Insulation resistance. Insulation resistance shall be 500 megohms minimum when tested as specified in 4.5.2.9.

3.6.15 Output ammeter. The output ammeter shall have a range of 0 to 5 amperes.

3.6.16 Output voltage meter. The output voltage meter shall have a range of 0 to 50 volts dc.

3.7 Construction

3.7.1 Case. The case shall be a ventilated steel case, dripproof 15° protected in accordance with MIL-STD-108 and shall not exceed 9 x 10 x 15 inches and finished in black vinyl enamel.

3.7.1.1 Front panel. The front panel shall be anodized aluminum in accordance with MIL-A-8625 with etched black letters. An ammeter of 0 to 5 amps range and a voltmeter of 0 to 50 volts range shall be mounted on the front panel. Figure 1 may be used as a guide.

3.7.1.2 Panel controls, meters and protective devices. The front panel shall include the following:

(a) Power on-off switch or circuit breaker
(b) Power-on indication light
(c) Output voltage controls

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(1) coarse
(2) fine

(d) Output current controls

(1) coarse
(2) fine

(e) Output ammeter
(f) Output voltmeter
(g) Over-voltage indicator light
(h) Output terminals shall provide a positive output, negative output and chassis ground.

3.7.2 Rear of case

3.7.2.1 The rear of the case shall include the following:

(a) Input line fuse
(b) Input terminals
(c) Output terminals shall provide a positive output, negative output and chassis ground.

3.7.2.2 The rear of the case shall contain the input terminals and line fuse holder. The input line fuse shall be in an indicating fuse holder.

3.8 Weight. The weight of the power supply shall be not greater than 50 pounds.

3.9 Marking. The power supply shall be marked in accordance with MIL-STD-1285 and MIL-STD-130. Markings shall remain legible after all tests.

3.10 Workmanship. Workmanship shall be in accordance with requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may use his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspections specified herein are classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

1. Inspection of preparation for delivery (see 4.4.2)

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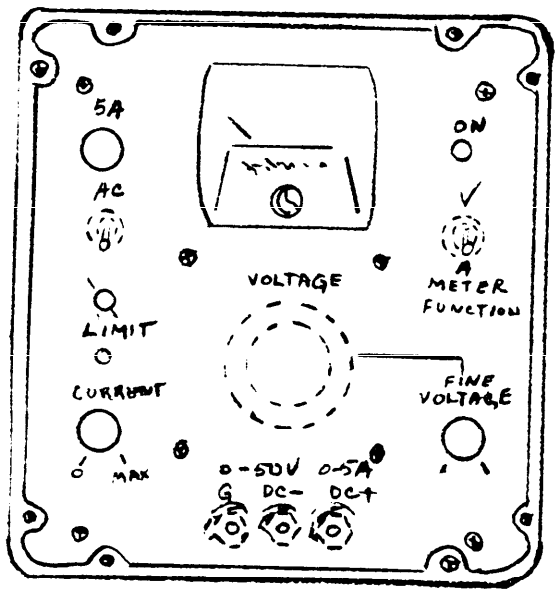


Figure 1. Front panel

NOTE: For similarity of mechanical contruction

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4.3 First article inspection. Unless otherwise specified (see 6.2), 2 units shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in table I. The first article inspection shall be witnessed and approved by the command or agency concerned.

Table I. First article inspection

Examination	Requirement	Method
Visual & Mechanical inspection (external)	3.7, 3.8, 3.9, 3.10	
Output voltage	3.6.2	4.5.2.2
Output current	3.6.3	4.5.2.1
Power drain	3.6.4	4.5.2.1
Voltage ripple	3.6.9	4.5.2.3
Output impedance	3.6.10	4.5.2.4
Overload	3.6.11	4.5.2.10
Maximum Operating Temperature (Heat run)	3.5.3.1	4.5.1.3.1
Run-in	3.6.13	4.5.2.8
Insulation resistance	3.6.14	4.5.2.9
Shock	3.5.1	4.5.1.1
Vibration	3.5.2	4.5.1.2

4.4 Quality conformance inspection.

4.4.1 Quality conformance inspection. Quality conformance inspection shall be made on every power supply offered for delivery. This inspection shall consist of the examinations and tests specified in table II.

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Table II. Quality conformance inspection

Examination	Requirement	Method
Visual & Mechanical inspection (external)	3.7, 3.8, 3.9, 3.10	
Output voltage	3.6.2	4.5.2.2
Output current	3.6.3	4.5.2.1
Power drain	3.6.4	4.5.2.1
Voltage ripple	3.6.9	4.5.2.3
Output impedance	3.6.10	4.5.2.4
Overload	3.6.11	4.5.2.10
Run-in	3.6.13	4.5.2.8
Insulation resistance	3.6.14	4.5.2.9

4.4.2 Inspection of preparation for delivery. Sample packages or packs and the inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5.

4.5 Test methods and procedures.

4.5.1 Environmental.

4.5.1.1 Shock. The power supply shall be subjected to a shock test in accordance with MIL-STD-202, Method 207. The power supply shall be mounted to the platform of the machine by use of canvas straps.

4.5.1.2 Vibration. The power supply shall be subjected to the vibration test in accordance with MIL-STD-202, Method 201. The power supply shall be attached to the vibration platform by use of canvas straps. The output voltage shall be monitored during the test.

4.5.1.3 Temperature.

4.5.1.3.1 Maximum operating temperature (heat run). Place the power supply in a chamber at 50° C and allow it to stabilize. Upon stabilization measure the resistance of the magnetic winding. Operate the power supply into a full load of 50 Vdc, 5 amps for 8 hours at 125 volts and 57 Hz. At the end of 8 hours remove load and immediately measure resistance of windings. Compute the temperature rise.

4.5.1.3.2 Low temperature test. The power supply shall be operated for 8 hours at 0° C into an output load of 50 Vdc 5 amps. Subsequent to the test the power supply shall not be damaged or adversely affected and shall meet the requirements of 3.6.2 through 3.6.5 at least once at the beginning, middle and ending of the 8 hours.

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4.5.1.3.3 Storage test. The power supply shall be placed in a chamber at -20° C for 4 hours. After 4 hours at -20° C the power supply shall be removed from the chamber and measurement shall be made to determine conformance with 3.6.2 through 3.6.5. The power supply shall be placed in a chamber at +85° C for 4 hours. After 4 hours at +85° C the power supply shall be removed from the chamber and measurements shall be made to determine conformance to 3.6.2 through 3.6.5.

4.5.2 Electrical

4.5.2.1 Operating test. Input power of 105 to 125 volts, 57 to 63 Hz to the power supply. With appropriate load applied, adjust output voltage and current with panel controls so that panel meters read 10, 20, 30, 40 and 50 volts at 1, 2, 3, 4 and 5 amps, respectively. Attach 1/2 percent accuracy voltmeter and ammeter to output terminals. The reading on the panel meters shall be within ± 2 percent of the applicable voltage and current on the test meters. When the power supply is adjusted for a dc load of 50 volts, 5 amps determine the primary power drain (see 3.6.4).

4.5.2.2 Voltage regulation. Using a voltmeter having an accuracy of 1/2 percent, measure the output voltage at no load conditions and at full load conditions of 50 Vdc 5 amps and determine the percent change between no load and full load conditions.

4.5.2.3 Voltage ripple. At full load conditions of 50 Vdc 5 amps, place an oscilloscope across the output terminals and determine the voltage ripple in millivolts.

4.5.2.4 Output impedance. At nonoperating condition, place an impedance meter across the output terminals and determine the output impedance.

4.5.2.5 Line regulation. At full load conditions of 50 Vdc, 5 amps, place a voltmeter having an accuracy of 1/2 percent across the output terminals. Vary the input voltage from 105 to 125 Vac and determine variation of output voltage.

4.5.2.6 Stability. Operate the power supply at full rated load of 50 Vdc, 5 amps for 24 hours. After warm-up, but prior to initiation of the 24-hour test, measure output voltage across output terminals using 1/2 percent accuracy meter. After 24 hours measure output voltage across output terminals using 1/2 percent accuracy meter. Compute percent change in output voltage between voltage measured after warm-up and prior to 24-hour test and after 24-hour test.

4.5.2.7 Temperature coefficient of output voltage. Operate the power supply into a full rated load of 50 Vdc, 5 amps at 25° C and measure output voltage. Increase temperature of chamber to 50° C without adjustment of controls and measure output voltage. Determination of output voltage at each temperature shall be after the power supply has stabilized. Calculate percent changes of voltage per degree Centigrade.

4.5.2.8 Run-in test. The power supply shall be operated at full rated load of 50 Vdc, 5 amps for 48 hours at 25° C. Subsequent to the test the

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power supply shall be inspected for damage and measurements shall be made to determine conformance with 3.6.2 through 3.6.5.

4.5.2.9 Insulation resistance. Insulation resistance shall be conducted in accordance with MIL-STD-202, Method 302, test condition B. Measurements shall be made between terminals, and terminals to ground.

4.5.2.10 Overload. Apply a short circuit across output terminals when the power supply is being operated at full rated output of 50 volts and 5 amperes. Subsequent to the test the power supply shall be inspected for damage and measurements shall be made to determine conformance with 3.6.2 through 3.6.5.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation and packaging, packing, and marking. Unless otherwise specified, preparation for delivery shall be in accordance with MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use. The power supply is used to convert the primary power to dc voltages necessary to operate Navy electronic equipments.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Number of first article samples to be submitted if other than specified in 4.3.
(c) Marking, levels of preservation and packaging and packing required (see 5.1)

6.3 First article. Invitations for bids should provide that the Government reserves the right to waive the requirements for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Preparing activity:
Navy - EC

(Project 5820-N559)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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NAVAL ELECTRONIC SYSTEMS COMMAND
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